

GLM by genetic distance, space, and time

```
## Setting up the matrices:
spatial_distance <- as.matrix(geo_dist)
gen_distance <- as.matrix(euc_dists)

time_dists <- as.matrix(gen_dist(time_df, dist_type = "euclidean"))
# Used alglatr gen_dist() function to make
# time matrix with the time dataframe as input.
# The regular dist() function did not format the
# data the same way.

##### code from Dagilis
## Run a GLM with interaction effects:
time_space_model = glm(c(gen_distance) ~ c(spatial_distance)*c(time_dists))
summary(time_space_model)
```

summary output

```
Call:
glm(formula = c(gen_distance) ~ c(spatial_distance) * c(time_dists))

Coefficients:
              Estimate Std. Error
(Intercept)    18.32065    0.28850
c(spatial_distance)    0.33635    0.02433
c(time_dists)      1.99428    0.13934
c(spatial_distance):c(time_dists) -0.07008    0.01090
              t value Pr(>|t|)
(Intercept)     63.503  < 2e-16 ***
c(spatial_distance)  13.822  < 2e-16 ***
c(time_dists)     14.312  < 2e-16 ***
c(spatial_distance):c(time_dists)  -6.427 1.55e-10 ***
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Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 33.61917)

Null deviance: 109443  on 2499  degrees of freedom
Residual deviance:  83913  on 2496  degrees of freedom
AIC: 15888

Number of Fisher Scoring iterations: 2
```